

data

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outline

regular (not gis) data: xls, csv, etc

gis data (has shapes, can make a map from it): shp, kml, etc

Notebook: dive into thematic/choloropleth maps

join/merge

Notebook: join/merge

DATA SOURCES

ex from past

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what are data?

- dataset/dataframe = matrix/spreadsheet/2D object
- u/a: unit of analysis: what do you study?
- u/a=# of obs=# of rows=sample size
- dataset has variables, which are the *attributes* of u/as
- say students: age; counties: water area
- cols=vars, rows=obs; vars characteristics of obs
- if several layers: may have several u/as
- eg counties: #18; hospitals:#700; ex of attr?

storage type: num (float,int) v str (object)

- strings are safer;
- eg str "08121" into num is "8121", a mistake!
- still, need to make str into num to do the math/map
- be careful, triple check, often problems and non-intuitive

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files

- .shp (along with bunch of others)
- .kml
- .json .geojson
- .gpkg
- more later in i/o sec in notebook <https://colab.research.google.com/github/theaok/gisPy/blob/main/map.ipynb#scrollTo=nzy1LGMMo7t4>

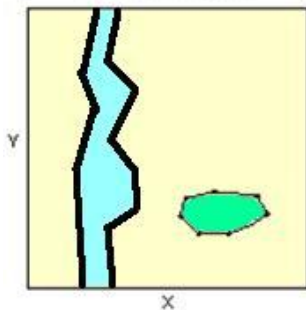
raster (picture) v vector (point, line, or polygon)

- raster (has resolution)
 - area covered by cells/pixels
 - each cell/pixel have values/colors
- vector (no resolution): all real world features:
 - points (dots/nodes): airports, cities, trees
 - lines (arcs): rivers, roads
 - polygons (areas): counties, cities

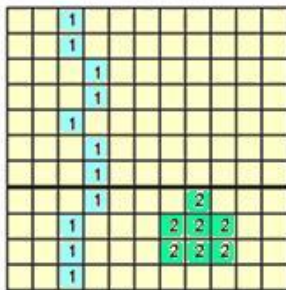
raster and vector



VECTORIAL



RASTER



gis data: layers of shapes with regular data

- data organized by layers
 - eg adm boundaries, roads; eg goog maps
- each layer: loc info (shapes)+often some regular data
 - data table with loc (shapes) must underlie a map
 - (the data table often has some regular data, too)
 - shapes=coords or lat/lon or x/y
 - thematic/choloropleth maps use different symbols/colors (themes) to show variation in regular data

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some real skills

- this is where the real value come from:
 - to bring different vars together to produce new insight
- if you just map vars from same or similar data:
 - it has probably already been done!
 - just goog: “what you study, map” and see images
- but combining creatively variety of vars:
 - there is no such map in the world!
 - eg https://scholarship.libraries.rutgers.edu/view/delivery/01RUT_INST/12643382240004646/13643522850004646

howto map regular (eg xls) data?

- it would likely have geo id:
 - hospital name/code, county name/id, etc
 - codes/ids are great: unique! (as opposed to names)
 - then google a shapefile that you can join with your data
- google “geo in you data, shapefile”
 - eg “NJ counties, shapefile”
- and then join the two to produce a map

the join problems

- “Camden county” $\neq$ “Camden”
- “Congo” $\neq$ “Congo, Republic of”
- “Great Britain” $\neq$ “United Kingdom”
- “Camden” $\neq$ “CAMDEN”
- “Camden ” $\neq$ “Camden” (space is a character !)
- “08012” $\neq$ “8012”
- be very careful; check the tables to see if it merged right
- does it make sense?
 - Camden richer than Cherry Hill?
 - the US poorer than India?

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don't trust anybody! neither yourself

- remember, always be critical
- compare with other source, eg goog 'nj counties income map'
- triangulate (measure concept differently):
 - <https://researchmethod.net/triangulation/>
 - <https://conjointly.com/kb/measurement-error/>
 - just goog picture, eg 'nj counties property values map'
 - looks about right
 - (other definition of the prices, but correlation is important)
- show to others, ask for comments

triple check

- merging (joining) data is tedious and tricky
- be careful, double, triple check
- easy to make mistake

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data management takes time! value your time!

- producing maps fast; data management 50-95% of time
 - figuring out, understanding, cleaning, documenting, combining, etc
- so we start with data management like join/merge
- spend time on data you care about and will use in your career!
- otherwise you'll waste 50+ hours

dataset of the week

- Google's open bldgs

<https://sites.research.google/gr/open-buildings/>

<https://gis.harvard.edu/event/>

abcd-gis-geography-colloquium-novel-geospatial-dataset-google-research

data ideas

- <https://www.dvrpc.org/data/>
- camden county <https://camdencountynj-ccdpc.opendata.arcgis.com/search?collection=Dataset> eg camden zoning :)
- NJ <https://gisdata-njdep.opendata.arcgis.com>
- Philly <https://www.opendataphilly.org>
- a lot!: <http://geocommons.com/search.html>
- just search for what you are interested in, say 'road'
- <https://www.policymap.com/maps>
- \$ to download data, but click 'Source' and download by hand
- open gov, especially city data, just few examples
- <https://data.cityofchicago.org/> , <http://opencityapps.org/> ,
<http://www.opendataphilly.org/>

data ideas

- NJ parcels

`https://njgin.nj.gov/njgin/edata/parcels/#!/`

- `https://www.njmap2.com/parcels/parcels/`

- `https://www.arcgis.com/apps/webappviewer/index.html?id=`

`95e52d6d16ce454fa449169e80b8ab44`

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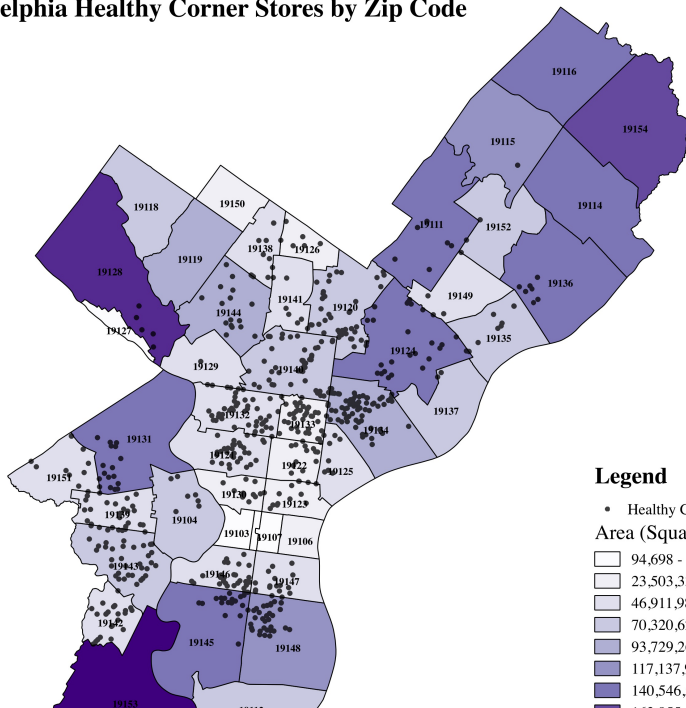
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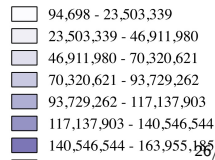
Philadelphia Healthy Corner Stores by Zip Code



Legend

- Healthy Corner Stores

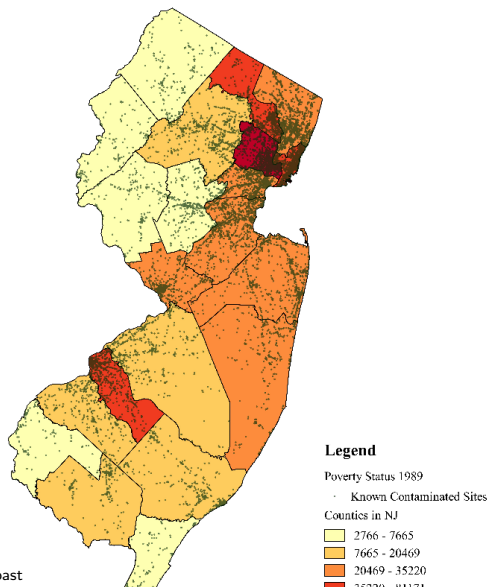
Area (Square Miles)



healthy corner stores

- makes sense to label zipcodes; right proportions
- these aren't sq miles! sq ft or meters!
- colors denote polygon sizes—so same info twice
- better could map educ, inc, age, bmi, etc
- dots could be little smaller or hollow so they overlap less
- make goog map and zoom in: show more detail
 - see environ: other businesses, pub transpo, sch, etc
- wonder about big healthy stores like wholefoods
- could denote big ones with big dots
- usually may want to put year on a map

Contaminations Sites in New Jersey 1992



contaminations

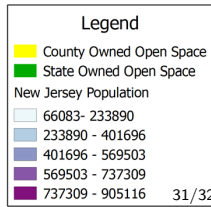
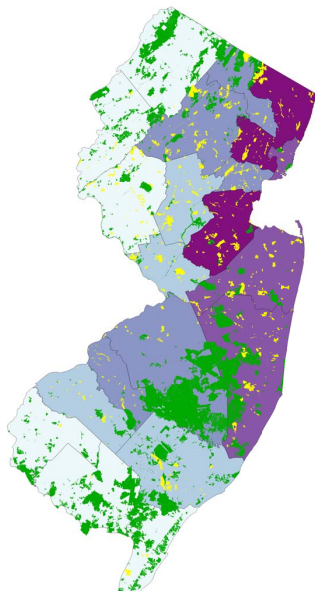
- perfect size and color for contaminated sites!
- doesn't overlap much but big enough to see
- and grayish good for contamination
- informative— NYC and Philly the worst
- excellent idea to relate poverty to contamination
- there is lit linking them! so nice test! [also can do race]
- could do poverty at municipal or census tract levels
- use space better! NJ should be bigger like Philly stores
- thousands must be set off by commas in legend
- very good to match contaminations and poverty by year!
- “poverty status”—guess counts; better %
- as in Philly map: zoom to Camden, have goog map in

contaminations

- http://www.nytimes.com/interactive/2015/07/08/us/census-race-map.html?_r=0
- in couple classes we'll be making online maps like this
- but already now you can do sth similar
- see footnote: census and socialexplorer.com: download data
- map in qgis and bring in background from googmaps
- with openlayers plugin

open space

New Jersey Preserved Open Space



open space

- excellent idea for map—open space related to population
- great use of multiple layers
- great non-cluttered borders
- can use space better—portrait orientation, bigger NJ
- use commas for population
- say for which year it is
- pop den probably more meaningful
- on the other hand, we already see size from map
- and so we can sort out density